

Lecture 18

Determinant $\rightarrow \det A = |A|$

$\rightarrow$ Matrix is invertible when determinant is non-zero.

Properties $\rightarrow$

① $\det I = 1$

② Exchanging rows reverses the sign of determinant. $\det P = \pm 1$

$$\begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1$$

$$\begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} = -1$$

③ $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = \underline{ad - bc}$

$$3a) \begin{vmatrix} ta & tb \\ c & d \end{vmatrix} = t \det \begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

$$3b) \begin{vmatrix} a+ta' & b+tb' \\ c & d \end{vmatrix} = \begin{vmatrix} a & b \\ c & d \end{vmatrix} + \begin{vmatrix} a' & b' \\ c & d \end{vmatrix}$$

- The determinant is a linear function of the first row if all other rows stay the same.

$$\det(A+B) \neq \det(A) + \det(B)$$

as it deals with all the row.

Exercise > What would be $\det(A+B)$

$$\det(A+B) = m \cdot \det(A) + \det \begin{pmatrix} -B_1 \\ -A_2 \\ \vdots \\ A_n \end{pmatrix} + \det \begin{pmatrix} A_1 \\ -B_2 \\ \vdots \end{pmatrix} \dots$$

④ If two rows are equal then
 $\det A = 0$

proof $\Rightarrow$ Let's say we exchange
those two rows.

$$\text{then } \det A = -\det A$$

$$\boxed{\det A = 0}$$

⑤ Subtract $k \times \text{row}_i$ from row_k

$\Rightarrow \det A$ does not change!

proof $\rightarrow \det A_{k-i} = \det A - k \det \underbrace{A_{k \leftrightarrow i}}_{\substack{\text{row } k \text{ row } i \\ k=i}}$

$$\boxed{\det A_{k-i} = \det A}$$

Since $k = i$
 $\det = 0$

⑥ Row of 0 leads to $\det A = 0$

proof: Substitute the row of zeros
by $a_0 - a_0$.

$$\det A_0 + (-1) (\det A_0) \\ \underline{\underline{= 0}}$$

$$\textcircled{7} \quad U = \begin{bmatrix} d_1 & x & x & x \\ & d_2 & x & x \\ & & \ddots & \vdots \\ 0 & & & d_n \end{bmatrix}$$

$$\boxed{\det U = d_1 \cdot d_2 \cdots d_n}$$

$\Rightarrow$ If we did some row exchanges
to get to this point then we
need to keep track of sign.

proof $\Rightarrow$ The determinant of a diagonal matrix is product of its diagonals.

$\Rightarrow$ Using property 3a $\rightarrow$

$$\det \begin{vmatrix} d_1 & & \\ & d_2 & \\ & & \ddots \\ & & & d_n \end{vmatrix} = d_1 \det \begin{vmatrix} & 0 & 0 & 0 \\ & d_2 & & \\ & & \ddots & \\ & & & d_n \end{vmatrix}$$

$$\approx d_1 \cdot d_2 \cdots d_n \det \begin{vmatrix} & & & \\ & 1 & & \\ & & \ddots & \\ & & & 1 \end{vmatrix} \approx$$

$\underbrace{\hspace{10em}}_{\rightarrow 1}$

$$\det \begin{vmatrix} d_1 & 0 & \\ & d_2 & \\ & & \ddots \\ & & & d_n \end{vmatrix} = d_1 \cdot d_2 \cdots d_n$$

⑧ $\det A = 0$ exactly when A is singular.

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} \rightarrow \begin{vmatrix} a & b \\ 0 & d - \frac{c}{a} \cdot b \end{vmatrix}$$

$$\det A = a \cdot \left(\frac{d - \frac{c}{a} \cdot b}{1} \right)$$

$$\det A = ad - bc$$

$$\textcircled{9} \quad \det AB = \det A \cdot \det B$$

$$\Rightarrow \det A^{-1} = \frac{1}{\det A}$$

$$\det (2A) = 2^{\overset{\text{rows/cols}}{m}} \det A$$

$$\det (cA) = c^n \det A$$

$$(10) \det(A^T) = \det A$$

$\Rightarrow$ proof is simple for diagonal matrix and we can reach a diagonal matrix with row ops & exchanges without changing the value of determinant so voila.

If a column is all zero, then also $\det A = 0$.

proof $A = LU$

$$A^T = U^T L^T$$

$$\begin{aligned} \frac{|U^T L^T|}{|LU|} &= \frac{|L^T| |U^T|}{|L| |U|} = \frac{|U^T|}{|U|} \end{aligned}$$

↑

$$|L| = |L^T| = 1$$

Another fact $\rightarrow$

If we get to a state after doing 7 row exchanges, we cannot $\checkmark$ get to the same state by doing 2, 4, 6, 8 -- row exchanges. $\checkmark$

Odd states can only be reached with odd row exchanges

$\Rightarrow$ Based on the sign of determinant.